

Evaluation Approach

How I would measure success

1. Task completion: the user successfully validates booking reference, view options, adds a luggage.
2. Successful escalation: conversations correctly escalated.
3. Correctness of API calls and API payload.
4. Speed to complete the task.
5. Number of agent-user exchanges needed to complete the add-luggage task (lower is generally better, as long as success and quality stay high).

Metrics I would track

1. Task completion rate: % of conversations that end with the user successfully adding luggage.
2. Safe escalation rate: % of conversations correctly escalated when automation cannot complete the request.
3. Automation rate: % of cases resolved without human intervention when they should be automatable.
4. Unknown answer rate: % of questions the agent was not able to answer to.
5. Number of agent-user exchanges needed to complete the add-luggage task (lower is generally better, as long as success and quality stay high).
6. Agent error log, agent log, API error log.
7. Measure time to resolution (for example, median time from first user message to successful luggage update).
8. API: Verify that the correct APIs are called with the correct payloads.

How would you identify areas for improvement?

1. Weekly review of escalated conversations, grouped by critical system error and agent error types.
2. Identify top failure patterns.
3. Run A/B tests on prompt/tool changes and keep only versions that improve completion rate, speed.
4. Deeply study the use case and domain (here, flight luggage management) to understand it end to end and identify valuable new features to ship.

How I would monitor quality over time

1. Define alerts for regressions: drop in completion rate, spike in escalations, spike in API/model errors.
2. Create automated tests that simulate user-agent conversations across different scenarios, with known and measurable success metrics, and run these tests on a regular schedule.
3. Periodically perform manual reviews of randomly selected conversations that were completed successfully.
4. Periodically run manual end-to-end tests of user-agent conversations across different scenarios.